Adopted Levels:tentative

Type	Author	History Citation	Literature Cutoff Date
Full Evaluation	Balraj Singh	NDS 156, 70 (2019)	31-Jan-2019

Q(β^-)=240 *CA*; S(n)=7320 *CA*; S(p)=4460 *CA*; Q(α)=8500 *CA* 1997Mo25
S(2n)=13350, S(2p)=7660 (theory,1997Mo25).

2004Og10: reinterpretation of one long-lived decay chain observed by 1999Og10 in ^{244}Pu(^{48}Ca,xn),E=236 MeV (E*=35 MeV) at Dubna, and tentatively assigned to the decay of ^{289}Fl formed via 3n-channel, as possibly originating from the decay of ^{290}Fl via the 2n-channel. The decay chain would be: ^{290}Fl → ^{286}Cn → ^{282}Ds → ^{278}Hs (decays by SF). The cross-section for this event was ≈0.2 pb. This event was not observed in later experiments done at energies of E*=41-53 MeV (2000Og05,2000Og07), as the α particle energy of the first decay was ≈0.1 MeV less than that attributed to ^{289}Fl in the later work. A 3-min α-SF event observed in 1999Og07 at E*=32.6 MeV for ^{290}Fl and assigned to ^{287}Fl was not seen by 2004Og10.

2016Ho09: further discussion of the event assigned to ^{289}Fl in 1999Og10, that the observed decay chain did not fit from energy point of view to any of the decay chains with N-Z ≤61, however it fitted well with the systematics if ^{290}Fl, instead of ^{289}Fl, was produced via 2n-channel in the reaction studied by 1999Og10. 2016Ho09 further surmise that decay chain ^{290}Fl → ^{286}Cn → ^{282}Ds → ^{278}Hs (decays by SF), as proposed earlier in 2004Og10, would be less likely as the SF half-life of 11.5 min *+550−52* for the decay of ^{278}Hs would be too long to be realistic. Thus the authors propose rare decay of ^{290}Fl by ε mode to ^{290}Nh, followed by ^{290}Nh → ^{286}Rg → ^{282}Mt → ^{278}Bh (decays by SF). For details see sections 4.7 and 5.1.2 in the text and Fig. 2 decay scheme.

2017Ka66: tentative assignment of ^{290}Fl from decay of ^{294}Lv produced in ^{248}Cm(^{48}Ca,2n),E=261.6 MeV (E^*=41.3 MeV reaction using RIKEN Linear Accelerator (RILAC) and gas-filled recoil ion separator (GARIS). One EVR-α-α-SF correlated event (chain #4) was interpreted in three ways, one being ^{294}Lv → ^{290}Fl → ^{286}Cn decay chain, while the other two involved (^{48}Ca,3n) channel producing ^{293}Lv → ^{289}Fl → ^{285}Cn, and ^{293}Lv → ^{289}Fl → ^{285}Cn → ^{281}Ds chains. See details of three interpretations of chain #4 in ^{294}Lv Adopted Levels dataset.

For theoretical studies, consult Nuclear Science References (NSR) database at NNDC, BNL for 64 primary references dealing with the half-lives and other aspects of nuclear structure in this mass region.

^{290}Fl Levels

E(level)	J^π	$T_{1/2}$	Comments
0?	0^+	19 s *+91−9*	%ε+%β^+≈100; %SF≤50 (2016Ho09); %α≤50 E(level): the reported activity is assumed to belong to the g.s. of ^{290}Fl. $T_{1/2}$: from 2016Ho09 review, also listed as 21 s *+101−10*. Theoretical $T_{1/2}(\alpha)$=33 h, $T_{1/2}(\beta)$>100 s (1997Mo25). Eα=9.710 MeV *45* (2016Ho09 review) from ^{290}Fl decay, 9.72 MeV *3* (2017Ka66).